CAS 3182 ARTIFICIAL INTELLIGENCE AND DECISION SUPPORT SYSTEMS

FINAL EXAM

29TH November 2024

TIME ALLOWED: THREE HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer Three (3) questions from this section.
3. This question paper carries a total of **100 marks.**
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted.

Question 1

Highland Secondary School is excited about the change in the school curriculum. To prepare their students early, they have decided to use technology to help implement a system that will enable them to learn and improve in different subjects. They have a busy school calendar every year, so they have decided to develop an intelligent tutoring system. They want an intelligent tutoring system designed as an expert system to help students learn a specific subject, such as mathematics. The system would adapt to the student's learning pace and provide personalized feedback and exercises. You have been hired as an AI analyst to help them with this project. They want you to explain the following:

a) Define what an expert system is. Explain the various stages involved in developing an Expert system. (10 marks)

b) How can expert systems be used to create a knowledge base of mathematical concepts and problem-solving strategies? Explain the role of knowledge representation and inference techniques in this context. (6 marks)

c) Discuss applying machine learning techniques, such as decision trees or neural networks, to analyze student performance data and adapt the tutoring strategies accordingly. (6 Marks)

d) How can natural language processing (NLP) techniques be used to enable natural language interaction between the student and the tutoring system? Consider the challenges and potential benefits of using NLP in this context. (7 marks)

e) Explain how reinforcement learning can be used to optimize the tutoring system's behavior. (6 marks)

f) Define the terms "Agents" and "Abstraction". (5 marks)

(Total 20 marks)

SECTION B: Attempt any THREE questions in this section

Question 2

Imagine an autonomous vehicle navigating a city. The vehicle must make real-time decisions to safely and efficiently reach its destination, considering traffic conditions, pedestrian movements, and road signs.

a) How can the autonomous vehicle be represented as an agent in the context of Artificial Intelligence? Define the key components of an agent, such as the perception, action, and decision-making processes, and explain how they apply to the autonomous vehicle.

(5marks)

b) Discuss the concept of abstraction in the context of autonomous vehicle navigation. How can abstraction be used to simplify the complex problem of navigating a city? Provide examples of abstract representations that could be used. (9marks)

c) How can machine learning techniques be used to learn and adapt to different driving scenarios? Explain the role of feature engineering and model selection in this context.

(6 marks)

(20 marks)

Question 3

a) Discuss the challenges of using traditional search algorithms like Breadth-First Search (BFS) and Depth-First Search (DFS) for autonomous vehicle navigation. What are the limitations of these algorithms in this context? (**Please refer to the scenario in question 2)** (5 marks)

b) Explain how informed search algorithms like A* search can be adapted for autonomous vehicle navigation. How can heuristics be designed to guide the search process efficiently?

(5 marks)

c) What are some of the key features that an autonomous vehicle's perception system should extract from its environment? How can these features be used to make informed decisions?

(5 marks)

d) Briefly explain the term" Forward Chaining "and "Backward Chaining" giving an example for each.

(5 marks)

(20 marks)

Question 4

UTH wants to develop a healthcare system that uses AI to diagnose diseases based on patient symptoms and medical test results. The aim is to provide accurate and timely diagnoses to improve patient outcomes.

a) How can the diagnosis system be represented as an agent? Define the key components of the agent, such as the perception, action, and decision-making processes (8 Marks)

b) Discuss the concept of abstraction in the context of medical diagnosis. How can medical data be abstracted to represent relevant information for diagnosis? Provide examples of abstract representations. (12 Marks)

(20 marks)

Question 5

Discuss the following terms

a) Explanation Facilities and Meta Knowledge (5 marks)

b) Fuzzy reasoning and Reasoning under uncertainty (5 marks)

c) Neural Networks and Deep Learning (5 marks)

d) DENDRAL and MYCIN (5 marks)

(20 marks)

END OF EXAM